VS Code GitHub Copilot — Interception Feasibility by Platform and License Tier

This document summarizes which GitHub/Copilot account configurations allow intercepting VS Code's built-in GitHub Copilot Chat LLM traffic for policy enforcement (routing it through our gateway, “Optimus”), across macOS and Windows. Three potential mechanisms were evaluated: a generic BaseUrl-style redirect, BYOK (Bring Your Own Key), and the github-enterprise.uri CAPI-discovery hijack mechanism.

The three levers, restated precisely

- **BaseUrl-style redirect** (what other agents like Claude use): closed on every tier, every platform. The extension's own overrideProxyUrl/overrideCapiUrl debug settings are confirmed ignored by the extension (per a filed GitHub issue), and HTTPS_PROXY/NODE_EXTRA_CA_CERTS support for chat/completion traffic is explicitly deferred per Microsoft's own design doc (agentHost/node/claude/CONTEXT.md). No license unlocks this — it is not a licensing gate, it is unimplemented/ignored by the extension itself.
- **BYOK (Bring Your Own Key):** works as a feature, but does not solve interception, on every tier, every platform. It is strictly additive — it registers a new selectable model next to Copilot's own default. It never redirects Copilot's default traffic, so it cannot be the enforcement mechanism regardless of license.
- **github-enterprise.uri discovery hijack:** the only lever that is actually license/account-tier-gated. VS Code's defaultAccount.ts derives its account-discovery URL from this setting, but only if the account's authentication provider is flagged enterprise: true. This requires a genuinely GHE-flavored sign-in — not just an Enterprise-tier license or a Copilot Business seat.

The matrix

#	Platform	Account Tier	BaseUrl	BYOK	github-enterprise.uri	License Needed	Needs Network-Egress Hook?
1	Mac	GitHub Consumer	No	Additive only	No (enterprise: false, confirmed)	None unlocks this	Yes — NETransparentProxyProvider, unbuilt
2	Mac	GitHub Enterprise (member, ordinary multi-tenant github.com — e.g. Zscaler's actual setup)	No	Additive only	No — tested live, stays false even with Enterprise membership	None unlocks this	Yes
3	Mac	GitHub Business (Copilot Business seat)	No	Additive only	No — tested live, Business-seat SKU confirmed, gate still false	None unlocks this	Yes
4	Mac	xxx.ghe.com (dedicated GHE Cloud instance)	No	Additive only	Likely yes — mechanism built and proven correct end-to-end against real traffic; never confirmed whether a real dedicated-instance sign-in actually sets enterprise: true	GitHub Enterprise Cloud with a dedicated instance (not Business, not ordinary Enterprise membership)	No — if confirmed
5	Windows	GitHub Consumer	No	Additive only	No (same code path; not independently tested on Windows but no platform-specific reason to differ)	None unlocks this	Yes — WFP redirect callout, unbuilt
6	Windows	GitHub Enterprise	No	Additive only	No	None unlocks this	Yes
7	Windows	GitHub Business	No	Additive only	No	None unlocks this	Yes
8	Windows	xxx.ghe.com	No	Additive only	Likely yes, same caveat as row 4	GitHub Enterprise Cloud with a dedicated instance	No — if confirmed

“GitHub Enterprise” vs “Copilot Business/Enterprise” — two different axes

Worth spelling out precisely, since conflating these two is an easy trap and is exactly what caused confusion mid-investigation. These are two independent axes, not one tier ladder:

- **Axis 1 — GitHub's own hosting/account plan** (governs SSO/SAML, audit logs, advanced security, and how the org is hosted): GitHub Free/Team (ordinary multi-tenant github.com), or **GitHub Enterprise** (Cloud or Server), which adds SSO/SAML, audit logs, policy enforcement, advanced security. Enterprise Cloud can be ordinary multi-tenant hosting (what Zscaler has) or a dedicated <slug>.ghe.com instance (what Zscaler confirmed it does *not* have).
- **Axis 2 — GitHub Copilot's own subscription tier** (a separate product, billed per seat, layered on top of whichever hosting plan the org has): Copilot Individual (personal), **Copilot Business** ($19/user/month — centralized seat management, policy toggles, content exclusions), or **Copilot Enterprise** ($39/user/month — higher AI-request allowance, earlier access to new models, deeper customization).

Per GitHub's own docs, Business and Enterprise are “the two Copilot plans for customers on GitHub Enterprise Cloud” — an enterprise account picks one per organization. Worth verifying rather than assuming: whether a standalone non-Enterprise org (plain GitHub Team) can still buy Copilot Business on its own — the docs found so far are scoped to the enterprise-account purchasing flow specifically, so this isn't confirmed either way from them alone.

Why this matters here: Zscaler's actual setup — real Enterprise-account membership (Axis 1, ordinary multi-tenant) plus a real Copilot Business seat (Axis 2) — is exactly the combination tested live, and enterprise: true stayed false. VS Code's gate cares about how the sign-in happened (tied to Axis 1's hosting model), not which Copilot billing tier is active (Axis 2). A hypothetical upgrade to Copilot Enterprise alone, with no dedicated GHE.com instance, would very likely produce the same enterprise: false result — the SKU doesn't touch the sign-in flow at all. This is also why the matrix below treats “License Needed” as a hosting-model question (dedicated GHE.com instance), never a Copilot-tier question.

What this means for licensing

Only one license tier currently matters, and it is narrow: GitHub Enterprise Cloud provisioned as a dedicated GHE.com instance. Not Copilot Business, not Copilot Enterprise as an add-on, not plain Enterprise-account membership — this exact combination (real Business seat + real Enterprise membership on ordinary github.com) was tested live and the gate stayed closed. A customer would need their GitHub org itself hosted on a dedicated <slug>.ghe.com instance, which is a GitHub hosting decision, separate from which Copilot plan they buy.

One open gap worth flagging: GitHub Enterprise Managed Users (EMU) is a candidate additional tier — it runs on ordinary github.com (no dedicated instance needed) but has its own distinct SSO-only sign-in flow, which could plausibly set enterprise: true without requiring GHE.com. This is an open question, currently pending confirmation from a customer's GitHub admin.

For every other combination — which represents most real customers (rows 1–3 and 5–7, six of eight rows) — there is currently no license that helps. The only path forward is a network-egress redirect fallback, which is genuinely unbuilt on both platforms, not merely ungated by license:

- **Mac:** NETransparentProxyProvider — a new Network Extension target, requiring its own entitlement; crash blast radius is limited to the one extension process.
- **Windows:** a WFP (Windows Filtering Platform) redirect callout — a new kernel-mode component requiring its own signing path (attestation vs. full WHQL); crash blast radius is the whole machine (bluescreen), a materially higher-stakes build than the Mac side.

Neither is committed engineering work today — both are “socialize, not build” per the existing team decision on the analogous Copilot-desktop-app case.

Summary

- Rows 1–3 and 5–7 (6 of 8): no license helps; requires unbuilt network-egress hook work.
- Rows 4 and 8: one specific license (dedicated GHE.com hosting) may close the gap, pending confirmation that a genuine GHE.com sign-in actually sets enterprise: true.
- Open item: confirm with customer GitHub admins whether EMU could be a lower-cost alternative path to the same enterprise: true gate.

Sources: Choosing your enterprise's plan for GitHub Copilot — GitHub Copilot Business vs Enterprise: Choosing the Right Plan for Your Team